

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper No. CHB-04A: Ancillary Chemistry-II (Basic Aspects of Chemistry)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following:

[5 × 2 = 10]

- (a) Discuss in brief the advantages of hydrogen-hydride fuel over conventional fossil fuels.
- (b) Write the names of the factors that affect the rate of photosynthesis in plants.
- (c) Differentiate between a co-factor and a prosthetic group.
- (d) Write the name of any one molecular disease and discuss its reason in brief.
- (e) Define the term Respiratory Quotient (RQ).

Question 2. Differentiate between the following pairs:

[7 × 2 = 14]

- (a) Aerobic respiration and anaerobic respiration
- (b) Isothermal thermodynamic processes and adiabatic processes
- (c) Covalent bonds and coordinate covalent bonds
- (d) Extensive thermodynamic properties and intensive properties

Question 3.

- (a) Discuss similarity and differences between haemoglobin and myoglobin. Explain the role of haemoglobin in the transport of O₂ and CO₂ in human blood. [8]
- (b) Discuss the 'light and dark reactions' of photosynthesis in brief. Name the substances which are produced in light reactions and used as inputs in dark reactions. [6]

Question 4.

- (a) What do you mean by Nylons? Discuss the reactions involved in the formation of Nylon-6,6 and Nylon-6. Give their uses. [8]
- (b) Discuss the energetics (Born-Haber cycle) for the formation of solid NaCl from solid sodium and chlorine gas. [6]

Question 5.

- (a) Discuss the composition, preparation, and properties of Bakelite and Polyneoprene. [6]
- (b) Explain the mechanism of metallic corrosion. How can it be prevented by sacrificial anode protection? [6]
- (c) Differentiate between fibrous and globular proteins. [2]